Student Details

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- Internship: Alfido Tech
- Project Title: Deploying Nginx using Docker on AWS EC2

Tools & Services Used

- AWS EC2
- Docker
- Nginx
- Ubuntu

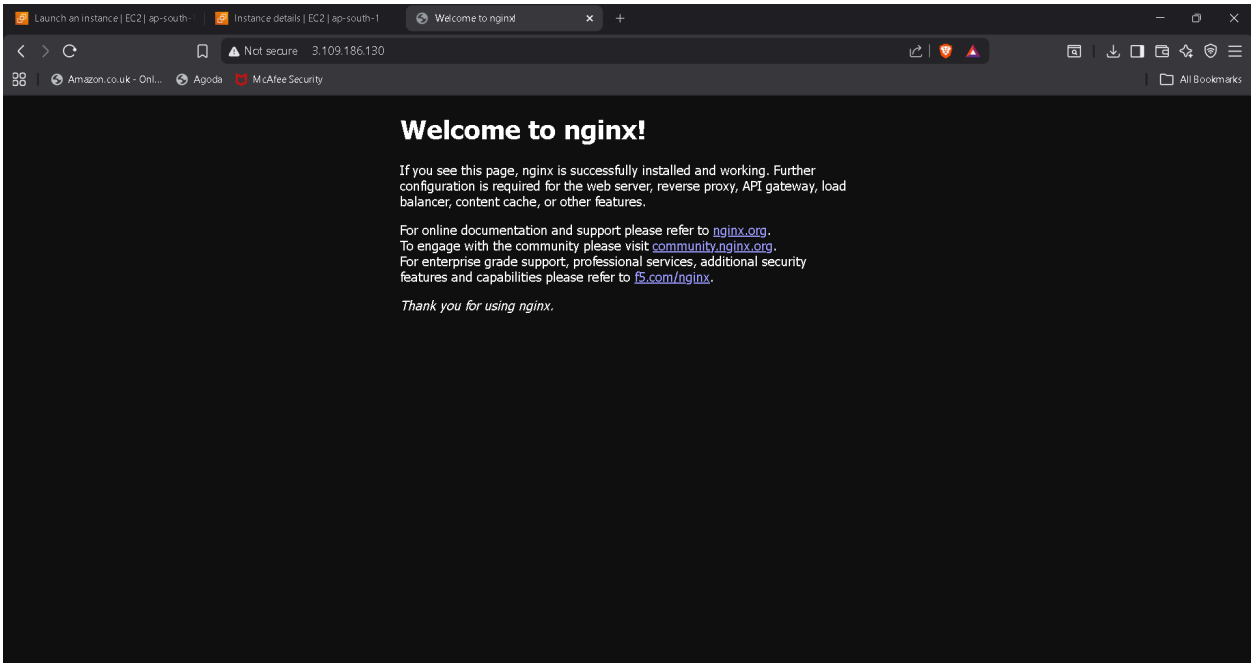
EC2 Setup

- Launched an EC2 instance
- Selected Ubuntu as the operating system
- Configured the security group to allow HTTP (Port 80) access**Docker Install**

- **Docker Install**

```
ubuntu@ip-172-31-42-53: ~
ubuntu@ip-172-31-42-53:~$ sudo apt update
sudo apt install docker.io -y
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1842 kB]
Get:14 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:15 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:16 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:17 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1570 kB]
Get:18 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [320 kB]
Get:19 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:20 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [33.0 kB]
Get:21 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2825 kB]
Get:22 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [655 kB]
Get:23 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:24 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.2 kB]
Get:25 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7520 B]
Get:26 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:27 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [500 B]
Get:28 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
Get:29 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9208 B]
Get:30 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7364 B]
Get:31 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:32 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.7 kB]
Get:33 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.2 kB]
Get:34 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.1 kB]
Get:35 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1480 B]
Get:36 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:37 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:38 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [780 B]
Get:39 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-en [372 B]
```

- ## Docker Start & Enable

```
ubuntu@ip-172-31-42-53:~$ sudo systemctl start docker
sudo systemctl enable docker
ubuntu@ip-172-31-42-53:~$ sudo docker run -d -p 80:80 --name myweb nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
ec781dee3f47: Pull complete
980067d12da2: Pull complete
4174e33a2c9e: Pull complete
6b40784e4837: Pull complete
f0b77348d9b0: Pull complete
0289d65812c3: Pull complete
9baba07a35b6: Pull complete
Digest: sha256:dec7a90bd0973b076832dc56933fe876bc014929e14b4ec49923951405370112
Status: Downloaded newer image for nginx:latest
67bc5199f4a908159d5663bb446bc304ade41591cb51a3d10c4d2f959c68c21c
ubuntu@ip-172-31-42-53:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                   NAMES
67bc5199f4a9   nginx     "/docker-entrypoint.…"   11 seconds ago   Up 11 seconds   0.0.0.0:80->80/tcp, [::]:80->80/tcp   myweb
ubuntu@ip-172-31-42-53:~$
```

- ## Result Verification